Operational Research: Tutorial 2

Linear Programming 1

Question 6 is difficult; however, it will be the basis for discussion in week 3. We will discuss the answers in detail in lecture to introduce some notation and points. It is not expected that you should be necessarily able to answer this yet.

The more complicated questions on the tutorial are marked by:

* Medium

** Hard

Questions 3, 5 and 6 run in the browser on the course site (open a new notebook there, or use the lecture demo notebook as a starting point). Question 4 needs desktop Excel with the Solver add-in.

Exercises

1) **Graphically solve the following linear programming (LP) problem**

maximise $5x + 17y$

subject to:

$$2x + 3y \leq 60$$
$$x + y \leq 25$$
$$x, y \geq 0$$

2) **Express the LP problem in matrix form.**
3) **Implement the problem in Python and use the `linprog` method to confirm your solution.**

   Note that to maximise a function f you need to minimise $-f$

4) **Implement the problem in Excel to confirm your solution.**
5) **A firm manufactures 2 products, X and Y.** *

   a)

   - **Each one is processed by a single machine, machine A, which has a capacity of 100 hours per week.**
   - **It takes 5 hours to make product X and 2 hours to make product Y.**
   - **The profit made (per unit) for X and Y is £12 and £5 respectively.**

   How many X's and Y's should be made per week so that the largest profit is made?

   Investigate the following:

   b) The firm adds a third product Z which takes 3 hours to manufacture and has a profit of £8. The firm only has enough material a week to make 20 of product Z

   c) In addition to product Z, the firm reduces the time to produce product X to 4.5 hours What do you notice about the solution, why might this be a non-optimal solution?

6) N&F (Management Consultants) have been employed to help the County dieticians develop attractive, nutritious, and economical diets for the county meals service. They have been employed in response to the many complaints about the palatability of the meals provided by this service in recent times. **

   The meals provided must satisfy certain basic restrictions on vitamin and fat content. Table 1 gives the target quantities for a sample of 4 vitamins and one fat type.

The quantities of each nutrient contained in one unit of each of 10 foods and the cost and attractiveness of a unit of each food are given in Table 2.

You are required to:

a. **Obtain the optimal solution given that the aim is to minimise the cost of the diet.**
b. **Obtain the optimal solution given that the aim is to maximise the attractiveness of the diet.**
c. **Investigate the effects on these solutions if there are maximum and minimum restrictions on both fat and vitamin content where maximum content = minimum content + 20%.**

	Minimum	Maximum
Vitamin A	100	-
Vitamin B	120	-
Vitamin C	200	-
Vitamin D	80	-
Fat	-	300

Table 1

Food	Vitamin A	Vitamin B	Vitamin C	Vitamin D	Fat	Cost	Attractiveness
1	18	4	-	6	-	10	6
2	8	6	22	14	-	6	1
3	22	10	-	-	8	8	5
4	6	12	-	6	-	16	5
5	-	14	-	10	-	8	6
6	-	-	20	8	6	3	4
7	8	-	-	6	20	6	10
8	4	-	8	-	40	8	10
9	-	6	10	12	-	12	6
10	12	20	-	4	-	12	7

Table 2